In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

In [2]:
```
df=pd.read_csv(r"C:\Users\Prasanth Sundar\indian_roads_dataset.csv")
```

In [3]:
```
df.sample()
```

Out[3]:

	accident_id	city	state	latitude	longitude	date	time	hour
6550	6550	Kolkata	West Bengal	22.625977	88.223833	2022-04-07	5:00	5

1 rows × 24 columns

In [4]:
```
df.describe()
```

Out[4]:

	accident_id	latitude	longitude	hour	is_weekend	2(
count	20000.000000	20000.000000	20000.000000	20000.000000	20000.000000	
mean	9999.500000	20.389207	78.173330	11.487200	0.286150	
std	5773.647028	6.165791	4.485967	6.945563	0.451972	
min	0.000000	12.800172	72.700017	0.000000	0.000000	
25%	4999.750000	13.198653	73.997979	5.000000	0.000000	
50%	9999.500000	18.812008	77.297000	12.000000	0.000000	
75%	14999.250000	28.402467	80.111089	18.000000	1.000000	
max	19999.000000	30.799960	88.499861	23.000000	1.000000	

In [5]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
Data columns (total 24 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   accident_id        20000 non-null  int64
 1   city               20000 non-null  str
 2   state              20000 non-null  str
 3   latitude           20000 non-null  float64
 4   longitude          20000 non-null  float64
 5   date               20000 non-null  str
 6   time               20000 non-null  str
 7   hour               20000 non-null  int64
 8   day_of_week        20000 non-null  str
 9   is_weekend         20000 non-null  int64
 10  road_type          20000 non-null  str
 11  lanes              20000 non-null  int64
 12  traffic_signal     20000 non-null  int64
 13  weather            20000 non-null  str
 14  visibility         20000 non-null  str
 15  temperature        20000 non-null  int64
 16  traffic_density    20000 non-null  str
 17  cause              20000 non-null  str
 18  accident_severity  20000 non-null  str
 19  vehicles_involved  20000 non-null  int64
 20  casualties         20000 non-null  int64
 21  is_peak_hour       20000 non-null  int64
 22  festival           115 non-null    str
 23  risk_score         20000 non-null  float64
dtypes: float64(3), int64(9), str(12)
memory usage: 3.7 MB
```

In [7]:
```
df.isnull().sum()
```

Out[7]:
```
accident_id              0
city                     0
state                    0
latitude                 0
longitude                0
date                     0
time                     0
hour                     0
day_of_week              0
is_weekend               0
road_type                0
lanes                    0
traffic_signal           0
weather                  0
visibility               0
temperature              0
traffic_density          0
cause                    0
accident_severity        0
vehicles_involved        0
casualties               0
is_peak_hour             0
festival             19885
risk_score               0
dtype: int64
```

In [8]:
```
df.tail()
```

Out[8]:

	accident_id	city	state	latitude	longitude	date	time
19995	19995	Bangalore	Karnataka	13.092276	77.599571	2022-09-29	3:00
19996	19996	Chennai	Tamil Nadu	13.172928	80.157062	2023-11-25	1:00
19997	19997	Chennai	Tamil Nadu	12.997170	80.150724	2022-06-18	15:00
19998	19998	Kolkata	West Bengal	22.454882	88.322213	2023-03-12	4:00
19999	19999	Delhi	Delhi	28.510266	77.065301	2024-07-05	23:00

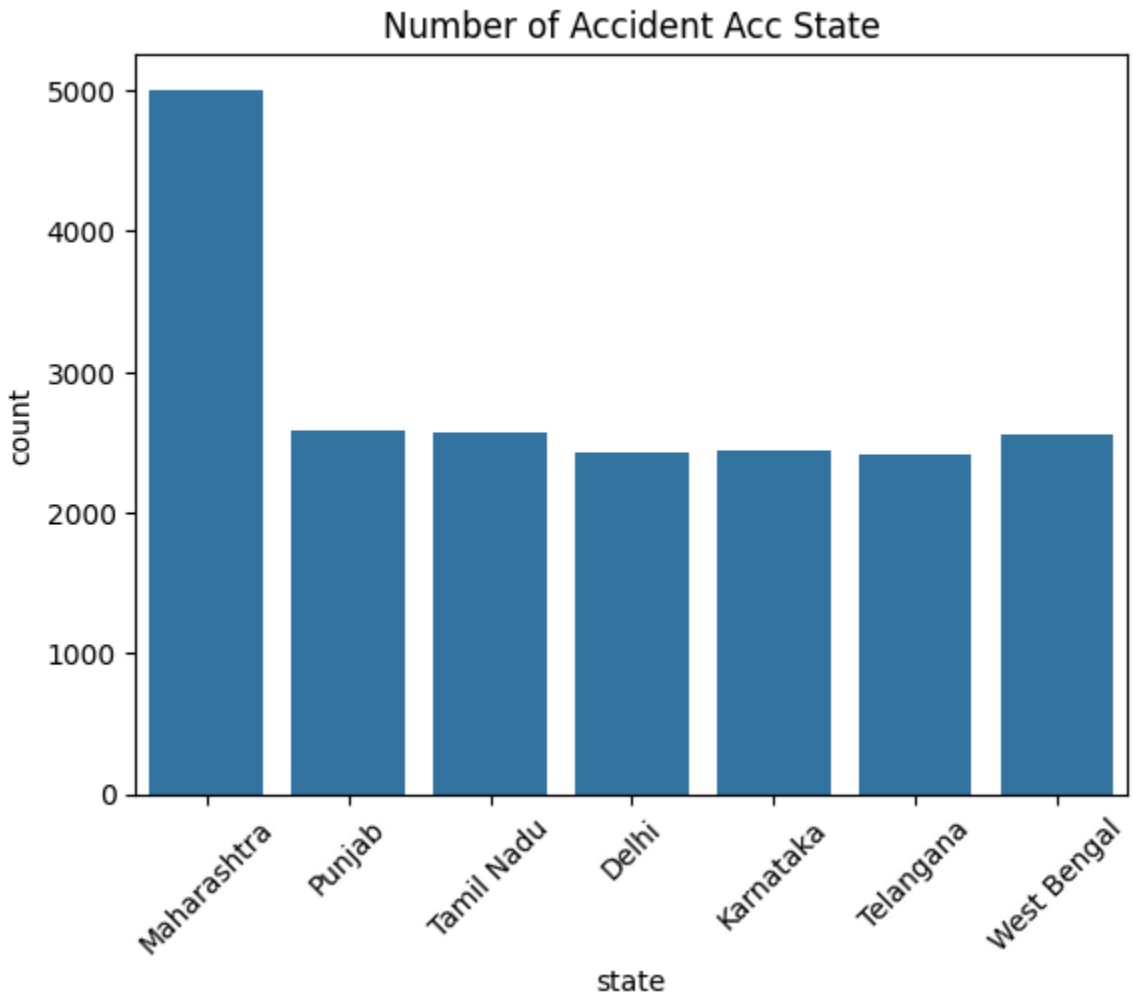
5 rows × 24 columns

In [9]:
```
df['city'].unique()
```

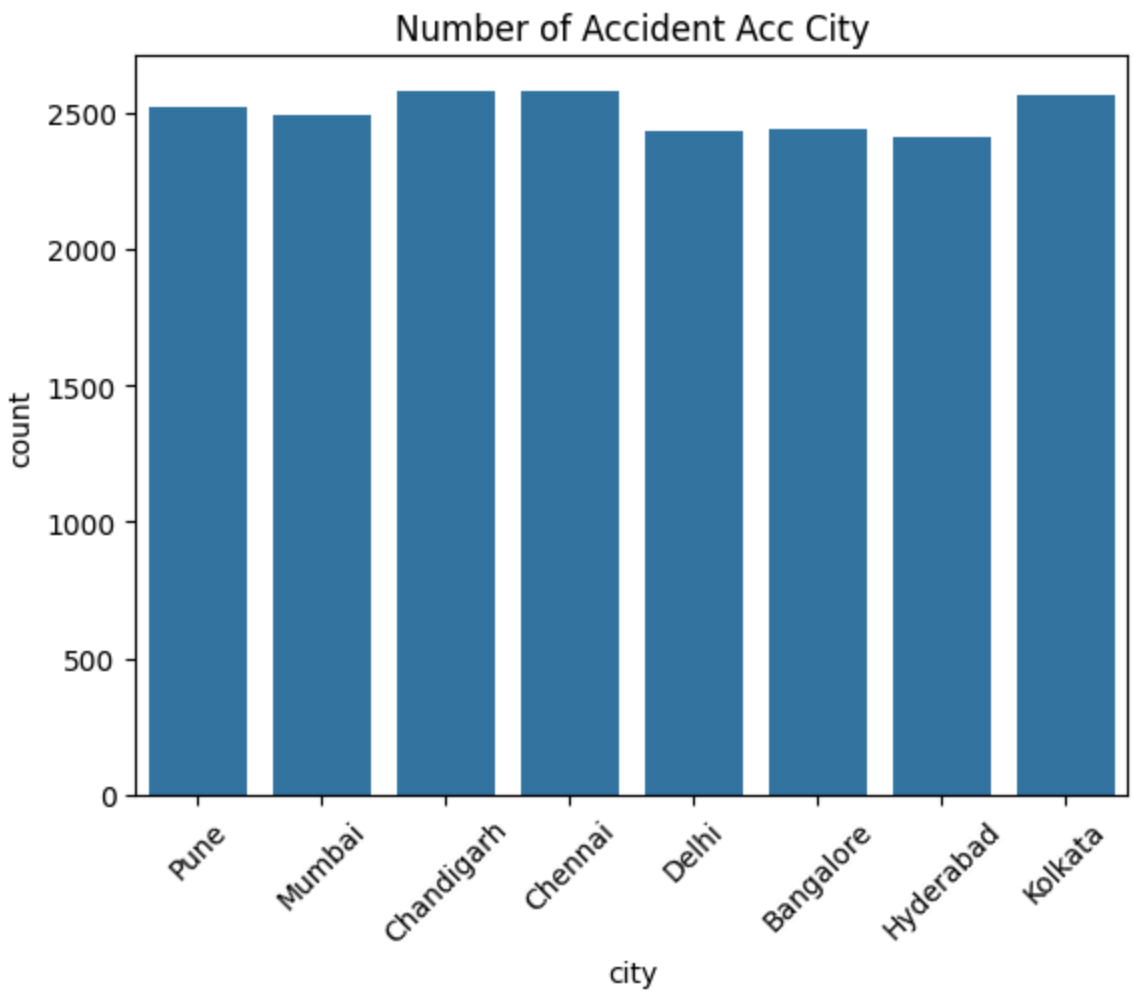
Out[9]:
```
<StringArray>
[      'Pune',     'Mumbai', 'Chandigarh',    'Chennai',      'Delhi',
  'Bangalore',  'Hyderabad',    'Kolkata']
Length: 8, dtype: str
```

In [10]:
```
sns.countplot(x = 'state', data = df)
plt.title('Number of Accident Acc State')
```

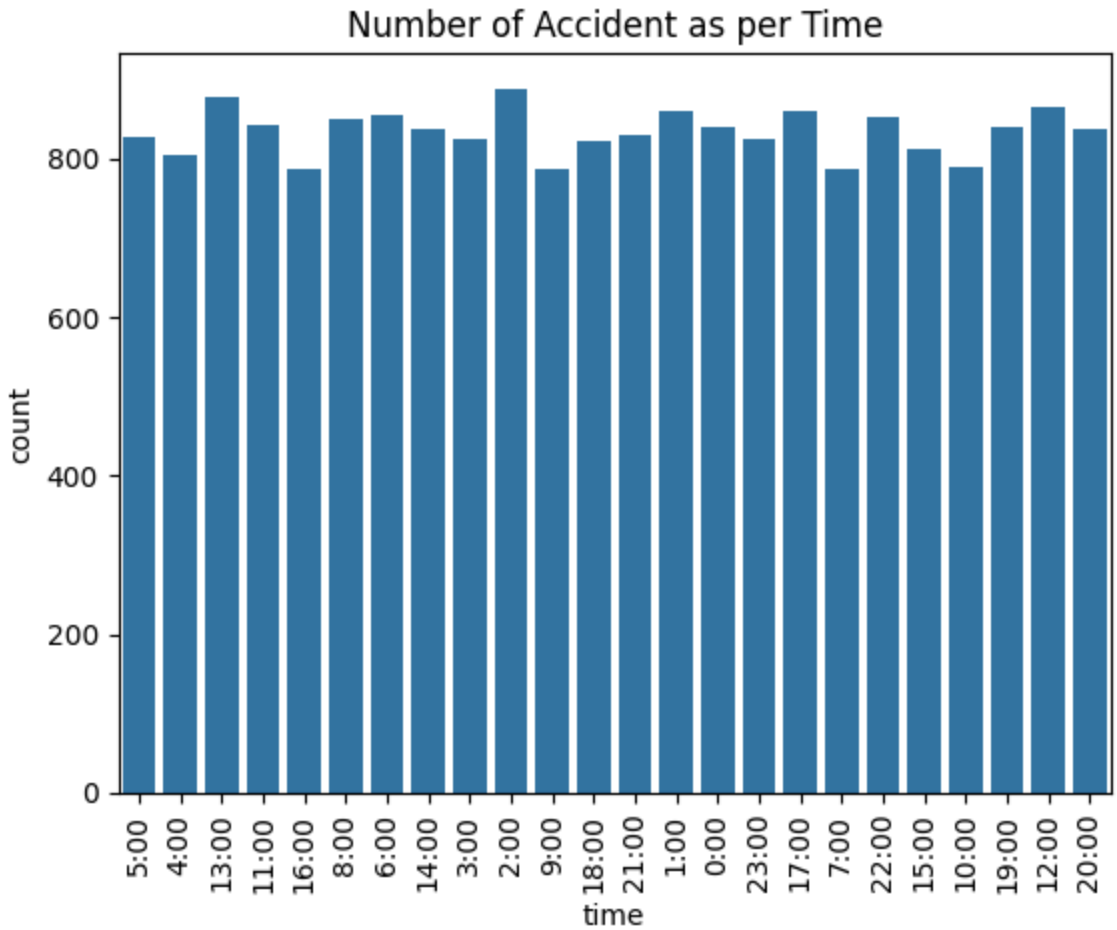
```
plt.xticks(rotation = 45)
plt.show()
```



In [11]:
```
sns.countplot(x = 'city', data = df)
plt.title('Number of Accident Acc City')
plt.xticks(rotation = 45)
plt.show()
```



In [12]:
```
sns.countplot(data=df, x='time')
plt.title('Number of Accident as per Time')
plt.xticks(rotation = 90)
plt.show()
```

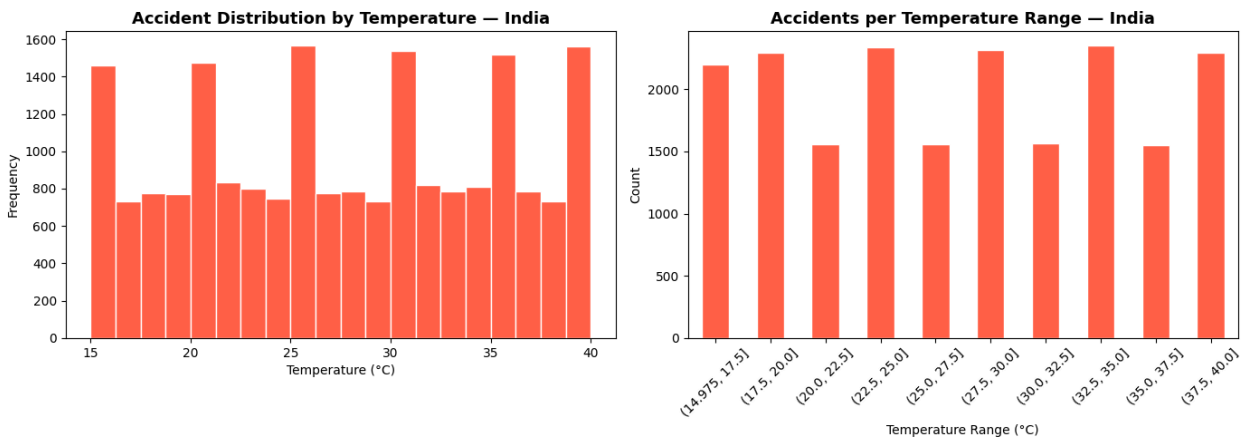


In [14]:
```
import warnings
warnings.filterwarnings("ignore")
```

In [15]:
```
print("=" * 55)
print("  ACCIDENTS BY TEMPERATURE — INDIA")
print("=" * 55)

temp_stats = df["temperature"].describe()
print("\n🌡️  Temperature stats (°C):\n", temp_stats.to_string())

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

df["temperature"].plot(kind="hist", bins=20, ax=axes[0],
                       color="tomato", edgecolor="white")
axes[0].set_title("Accident Distribution by Temperature — India",
                  fontsize=13, fontweight="bold")
axes[0].set_xlabel("Temperature (°C)"); axes[0].set_ylabel("Frequency")

# Average accidents per temperature bucket
df["temp_bin"] = pd.cut(df["temperature"], bins=10)
temp_grp = df.groupby("temp_bin", observed=True).size()
temp_grp.plot(kind="bar", ax=axes[1], color="tomato", edgecolor="white")
axes[1].set_title("Accidents per Temperature Range — India",
                  fontsize=13, fontweight="bold")
```

```
axes[1].set_xlabel("Temperature Range (°C)"); axes[1].set_ylabel("Count")
axes[1].tick_params(axis="x", rotation=45)

plt.tight_layout()
plt.savefig("india_accidents_by_temperature.png", dpi=150)
plt.show()
print("📊 Chart saved → india_accidents_by_temperature.png")
```

```
========================================================
  ACCIDENTS BY TEMPERATURE — INDIA
========================================================

🌡️  Temperature stats (°C):
 count    20000.000000
mean        27.579600
std          7.454746
min         15.000000
25%         21.000000
50%         28.000000
75%         34.000000
max         40.000000
```



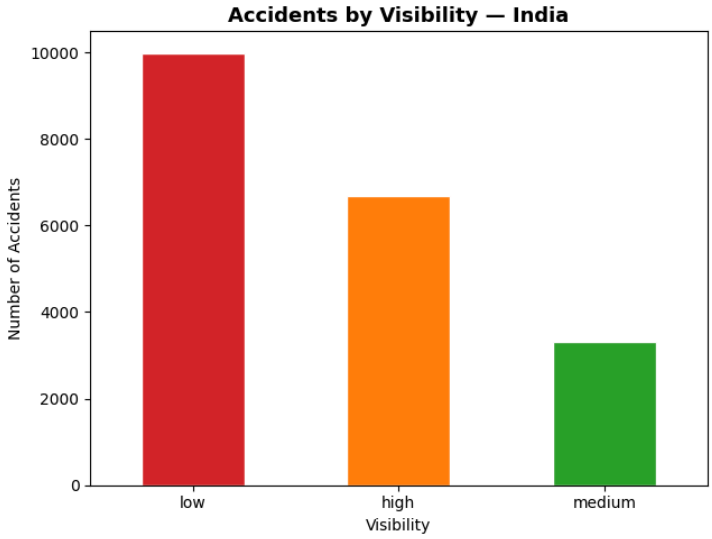

```
📊 Chart saved → india_accidents_by_temperature.png
```

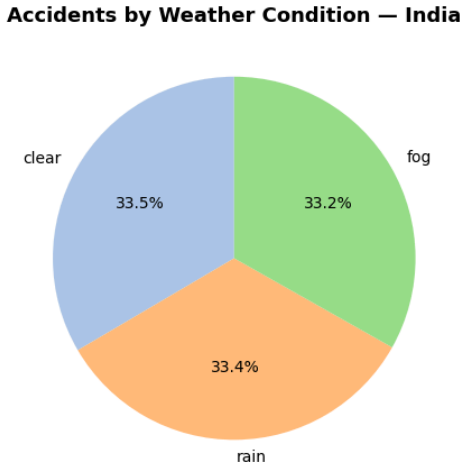
In [16]:

```
print("=" * 55)
print("  ACCIDENTS BY VISIBILITY — INDIA")
print("=" * 55)

by_vis = df.groupby("visibility").size().sort_values(ascending=False)
by_weather = df.groupby("weather").size().sort_values(ascending=False)

print("\n👁️  Accidents by Visibility level:")
print(by_vis.to_string())

print("\n⛅ Accidents by Weather:")
print(by_weather.to_string())

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

by_vis.plot(kind="bar", ax=axes[0], color=["#d62728", "#ff7f0e", "#2ca02c"],
            edgecolor="white")
axes[0].set_title("Accidents by Visibility — India", fontsize=13, fontweight="
```

```
axes[0].set_xlabel("Visibility"); axes[0].set_ylabel("Number of Accidents")
axes[0].tick_params(axis="x", rotation=0)

by_weather.plot(kind="pie", ax=axes[1], autopct="%1.1f%%",
                colors=["#aec7e8", "#ffbb78", "#98df8a"], startangle=90)
axes[1].set_title("Accidents by Weather Condition — India", fontsize=13, fontw
axes[1].set_ylabel("")

plt.tight_layout()
plt.savefig("india_accidents_by_visibility.png", dpi=150)
plt.show()
print("📊 Chart saved → india_accidents_by_visibility.png")
```

```
=======================================================
  ACCIDENTS BY VISIBILITY — INDIA
=======================================================

👁️  Accidents by Visibility level:
visibility
low       9987
high      6690
medium    3323

⛅ Accidents by Weather:
weather
clear    6690
rain     6677
fog      6633
```





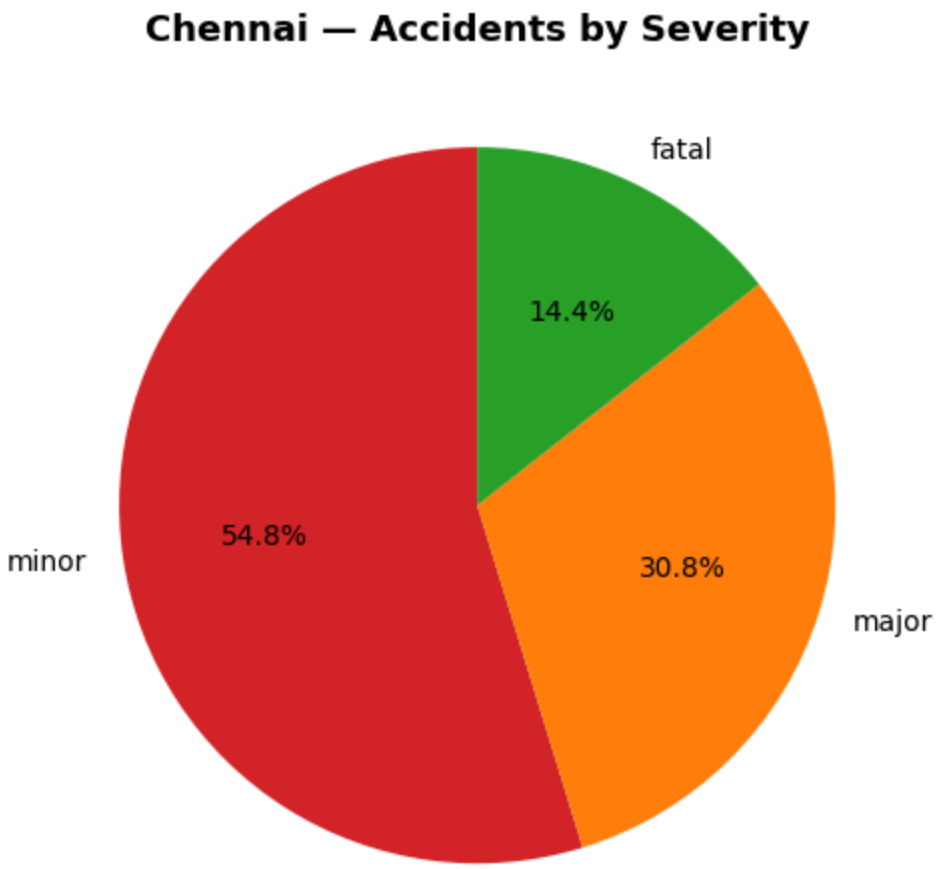

```
📊 Chart saved → india_accidents_by_visibility.png
```

In [25]:

```
chennai = df[df["city"] == "Chennai"].copy()
print(f"✅ Chennai rows: {len(chennai):,}")
chennai.head()


# ── CELL 8 : Chennai — Total Accidents Overview ──────────────
print("=" * 55)
print("  CHENNAI — TOTAL ACCIDENTS OVERVIEW")
```

```
print("=" * 55)

print(f"\n📍 Total Accidents in Chennai : {len(chennai):,}")
print(f"   Fatal   : {(chennai['accident_severity'].str.lower() == 'fatal').su
print(f"   Major   : {(chennai['accident_severity'].str.lower() == 'major').su
print(f"   Minor   : {(chennai['accident_severity'].str.lower() == 'minor').su

fig, ax = plt.subplots(figsize=(6, 5))
chennai["accident_severity"].value_counts().plot(
    kind="pie", ax=ax, autopct="%1.1f%%",
    colors=["#d62728", "#ff7f0e", "#2ca02c"], startangle=90
)
ax.set_title("Chennai — Accidents by Severity", fontsize=13, fontweight="bold"
ax.set_ylabel("")
plt.tight_layout()
plt.savefig("chennai_accident_severity.png", dpi=150)
plt.show()
print("📊 Chart saved → chennai_accident_severity.png")
```

```
✅ Chennai rows: 2,575
=======================================================
  CHENNAI — TOTAL ACCIDENTS OVERVIEW
=======================================================

📍 Total Accidents in Chennai : 2,575
   Fatal   : 371
   Major   : 794
   Minor   : 1410
```

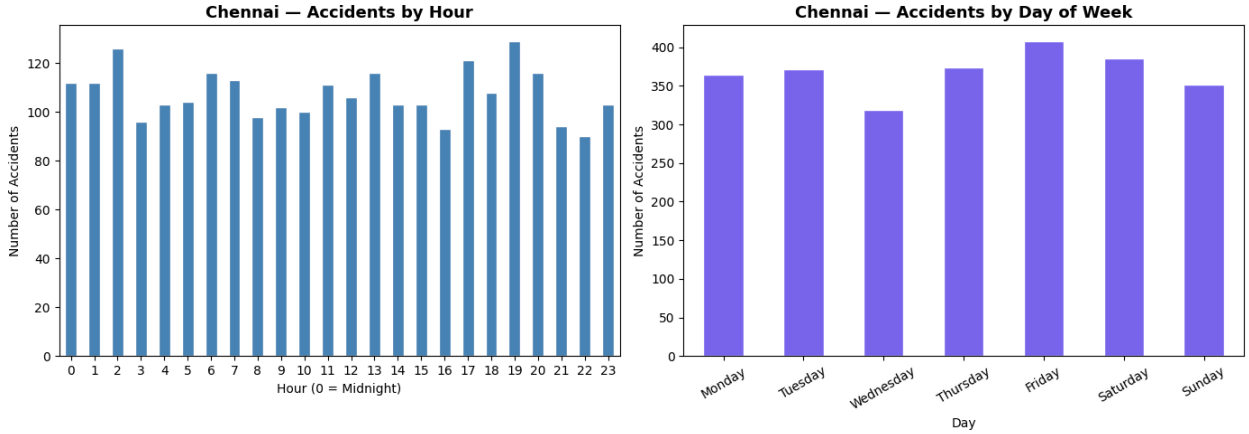


📊 Chart saved → chennai_accident_severity.png

In [26]:
```
print("=" * 55)
print("  CHENNAI — TIME OF ACCIDENT")
print("=" * 55)

chennai_hour = chennai.groupby("hour").size()
chennai_dow  = chennai.groupby("day_of_week").size()

print("\n⏰ Accidents per Hour (Chennai):")
print(chennai_hour.to_string())

print("\n📅 Accidents per Day of Week (Chennai):")
print(chennai_dow.to_string())

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

chennai_hour.plot(kind="bar", ax=axes[0], color="steelblue", edgecolor="white"
axes[0].set_title("Chennai — Accidents by Hour", fontsize=13, fontweight="bolc
axes[0].set_xlabel("Hour (0 = Midnight)"); axes[0].set_ylabel("Number of Accic
axes[0].tick_params(axis="x", rotation=0)

day_order = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday
chennai_dow.reindex(day_order, fill_value=0).plot(
    kind="bar", ax=axes[1], color="mediumslateblue", edgecolor="white"
```

```
)
axes[1].set_title("Chennai — Accidents by Day of Week", fontsize=13, fontweigh
axes[1].set_xlabel("Day"); axes[1].set_ylabel("Number of Accidents")
axes[1].tick_params(axis="x", rotation=30)

plt.tight_layout()
plt.savefig("chennai_accidents_by_time.png", dpi=150)
plt.show()
print("📊 Chart saved → chennai_accidents_by_time.png")
```

```
============================================================
  CHENNAI — TIME OF ACCIDENT
============================================================

⏰ Accidents per Hour (Chennai):
hour
0     112
1     112
2     126
3      96
4     103
5     104
6     116
7     113
8      98
9     102
10    100
11    111
12    106
13    116
14    103
15    103
16     93
17    121
18    108
19    129
20    116
21     94
22     90
23    103

📅 Accidents per Day of Week (Chennai):
day_of_week
Friday       408
Monday       365
Saturday     386
Sunday       351
Thursday     374
Tuesday      372
Wednesday    319
```

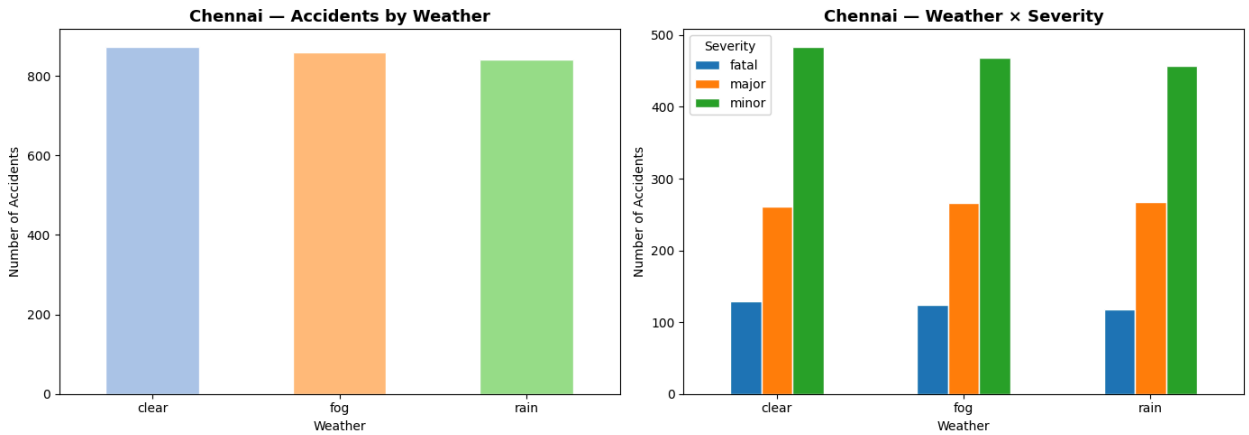



```
📊 Chart saved → chennai_accidents_by_time.png
```

In [27]:

```
print("=" * 55)
print("  CHENNAI — WEATHER CONDITIONS")
print("=" * 55)

ch_weather = chennai.groupby("weather").size().sort_values(ascending=False)
print("\n⛅ Accidents by Weather (Chennai):")
print(ch_weather.to_string())

# Cross-tab: weather × severity
ch_cross = pd.crosstab(chennai["weather"], chennai["accident_severity"])
print("\nWeather × Severity Cross-Tab:")
print(ch_cross)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

ch_weather.plot(kind="bar", ax=axes[0], color=["#aec7e8", "#ffbb78", "#98df8a"
                edgecolor="white")
axes[0].set_title("Chennai — Accidents by Weather", fontsize=13, fontweight="b
axes[0].set_xlabel("Weather"); axes[0].set_ylabel("Number of Accidents")
axes[0].tick_params(axis="x", rotation=0)

ch_cross.plot(kind="bar", ax=axes[1], edgecolor="white")
axes[1].set_title("Chennai — Weather × Severity", fontsize=13, fontweight="bol
axes[1].set_xlabel("Weather"); axes[1].set_ylabel("Number of Accidents")
axes[1].tick_params(axis="x", rotation=0)
axes[1].legend(title="Severity")

plt.tight_layout()
plt.savefig("chennai_accidents_by_weather.png", dpi=150)
plt.show()
print("📊 Chart saved → chennai_accidents_by_weather.png")
```

```
=======================================================
  CHENNAI — WEATHER CONDITIONS
=======================================================

⛅ Accidents by Weather (Chennai):
weather
clear    874
fog      859
rain     842

Weather × Severity Cross-Tab:
accident_severity  fatal  major  minor
weather
clear                129    261    484
fog                  124    266    469
rain                 118    267    457
```




```
📊 Chart saved → chennai_accidents_by_weather.png
```

In [32]:
```
chennai_hour = chennai.groupby("hour").size()
chennai_dow  = chennai.groupby("day_of_week").size()
ch_weather   = chennai.groupby("weather").size().sort_values(ascending=False)
ch_vis       = chennai.groupby("visibility").size().sort_values(ascending=Fals
```

In [33]:
```
print("=" * 55)
print("  CHENNAI — COMPLETE SUMMARY")
print("=" * 55)

# Recompute all dependencies so this cell runs standalone
chennai_hour = chennai.groupby("hour").size()
chennai_dow  = chennai.groupby("day_of_week").size()
ch_weather   = chennai.groupby("weather").size().sort_values(ascending=False)
ch_vis       = chennai.groupby("visibility").size().sort_values(ascending=Fals

summary_chennai = {
    "Total Accidents"          : len(chennai),
    "Peak Accident Hour"       : f"{int(chennai_hour.idxmax())}:00",
    "Busiest Day"              : chennai_dow.idxmax(),
    "Most Common Weather"      : ch_weather.idxmax(),
    "Highest Risk Visibility"  : ch_vis.idxmax(),
    "Fatal Accidents"          : (chennai["accident_severity"].str.lower() ==
```

```
    "Major Accidents"          : (chennai["accident_severity"].str.lower() ==
    "Minor Accidents"          : (chennai["accident_severity"].str.lower() ==
    "Avg Temperature (°C)"     : round(chennai["temperature"].mean(), 1),
    "Peak Hour Accidents"      : (chennai["is_peak_hour"] == 1).sum(),
    "Weekend Accidents"        : (chennai["is_weekend"] == 1).sum(),
}

for k, v in summary_chennai.items():
    print(f"  {k:<35}: {v}")
```

```
============================================================
  CHENNAI — COMPLETE SUMMARY
============================================================
  Total Accidents                    : 2575
  Peak Accident Hour                 : 19:00
  Busiest Day                        : Friday
  Most Common Weather                : clear
  Highest Risk Visibility            : low
  Fatal Accidents                    : 371
  Major Accidents                    : 794
  Minor Accidents                    : 1410
  Avg Temperature (°C)               : 27.6
  Peak Hour Accidents                : 658
  Weekend Accidents                  : 737
```

In [34]:

```
print("\n📋 Hour × Weather × Visibility Breakdown (Chennai):")
detailed = (
    chennai.groupby(["hour", "weather", "visibility"])
           .size()
           .reset_index(name="accident_count")
           .sort_values("accident_count", ascending=False)
)
print(detailed.head(20).to_string(index=False))
```

```
📋 Hour × Weather × Visibility Breakdown (Chennai):
 hour weather visibility  accident_count
   19   clear       high              53
   20   clear       high              52
    6   clear       high              49
   11   clear       high              48
    7     fog        low              47
    0     fog        low              47
   17     fog        low              46
    5     fog        low              43
   13     fog        low              43
   13   clear       high              41
    8     fog        low              41
   18     fog        low              40
    2     fog        low              40
    2   clear       high              40
   15     fog        low              39
   14     fog        low              39
   18   clear       high              39
    7   clear       high              38
   16   clear       high              37
   22   clear       high              37
```

In [40]:
```
print("=" * 55)
print("  CHENNAI — Ticket to HELL Period 💀 ")
print("=" * 55)
print("""
Quick findings from the data:
•        Chennai has 2,575 accidents — peak at 7 PM, worst day Friday
•        Fog + low visibility is the most dangerous combo in Chennai
•        Average temperature in Chennai accidents: 27.6°C """)
```

```
=======================================================
  CHENNAI — Ticket to HELL Period 💀
=======================================================

Quick findings from the data:
•        Chennai has 2,575 accidents — peak at 7 PM, worst day Friday
•        Fog + low visibility is the most dangerous combo in Chennai
•        Average temperature in Chennai accidents: 27.6°C
```

In [42]:
```
dataset = pd.get_dummies(df, drop_first = True)
```

In [43]:
```
df.shape
```

Out[43]:
```
(20000, 25)
```

In [44]:
```
X = dataset.drop('risk_score', axis = 1)
y = dataset['risk_score']
```

In [45]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, ran
```

In [46]:
```
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

models = {
    "Linear Regression": Pipeline([
        ("scaler", StandardScaler()),
        ("model", LinearRegression())
    ]),
    "Random Forest": RandomForestRegressor(n_estimators=100, random_state=42),
    "Gradient Boosting": GradientBoostingRegressor()
}
```

In [47]:
```
from sklearn.metrics import r2_score, mean_squared_error

results = {}

for name, model in models.items():

    model.fit(X_train, y_train)
    preds = model.predict(X_test)

    r2 = r2_score(y_test, preds)
    rmse = mean_squared_error(y_test, preds)

    results[name] = {"R2": r2, "RMSE": rmse}
    print(f"{name} → R2: {r2:.4f}, RMSE: {rmse:.4f}")
```

```
Linear Regression → R2: 1.0000, RMSE: 0.0000
Random Forest → R2: 0.9987, RMSE: 0.0001
Gradient Boosting → R2: 0.9995, RMSE: 0.0000
```

In [48]:
```
y_shuffled = np.random.permutation(y)

model.fit(X_train, y_shuffled[:len(X_train)])
preds = model.predict(X_test)

print("R2 after shuffle:", r2_score(y_test, preds))
```

```
R2 after shuffle: -5.9990916955854345e-05
```

In [49]:
```
corr = dataset.corr(numeric_only=True)["risk_score"].sort_values(ascending=Fal
print(corr.head(15))
```

```
risk_score         1.000000
visibility_low     0.694073
weather_fog        0.489390
is_peak_hour       0.436848
time_18:00         0.166946
time_8:00          0.164792
weather_rain       0.164097
time_17:00         0.162786
time_10:00         0.159789
time_19:00         0.147679
time_9:00          0.145881
hour               0.122017
casualties         0.119954
festival_Eid       0.031283
date_2023-04-22    0.025734
Name: risk_score, dtype: float64
```

In [54]:

```python
print("=" * 55)
print("""Author = Prasanth Sundar
DATA ANALYST
Linkedin =https://www.linkedin.com/in/prasanth-sundar-b65475359/
Github = https://github.com/Prashantsundar """)
print("=" * 55)
```

```
=======================================================
Author = Prasanth Sundar
DATA ANALYST
Linkedin =https://www.linkedin.com/in/prasanth-sundar-b65475359/
Github = https://github.com/Prashantsundar
=======================================================
```

In [ ]: